TABLEAU PORTFOLIO PROJECT

GROUP F: WORLD LIFE EXPECTANCY -- ANIMATED DATA STORY

World Life Expectancy by Fertility Rates

Animated scatter plot story with bubble sizing, regional color encoding, and year filter

Tableau | Animation | Scatter Plot | Bubble Chart | Story | Global Health | Fertility Rate | Pages Shelf

Project Summary

An animated Tableau Story scatter plot visualizing the relationship between Fertility Rate (x-axis) and Life Expectancy (y-axis) across ~180 countries, grouped by 7 world regions, with bubble size encoding population. The animated Pages shelf steps through years, revealing how the global fertility-life expectancy relationship has shifted over decades. Shown at Year 1967 with "Show history" trail feature enabled.

Thomas E. Foster

Senior Data Analyst | Tableau | Business Intelligence | Data Visualization

github.com/thomasfoster | thomasfoster79@yahoo.com | (540) 570-6094

Pea Ridge, AR 72751

1 PROJECT OVERVIEW

Workbook:	World Life Expectancy by Fertility Rates -- Story 2
Views:	1 animated scatter plot within a Tableau Story
Data Source:	World Bank / Gapminder-style dataset -- fertility rate, life expectancy, population by country/year
X-Axis:	Fertility Rate (avg. births per woman, 1.0 to 8.0+)
Y-Axis:	Life Expectancy (years, 30 to 85)
Bubble Size:	Population (country-level)
Color Encoding:	7 world regions (Sub-Saharan Africa, South Asia, North America, Middle East & North Africa, Latin America & Ca
Animation:	Pages shelf on [Year] -- steps through decades from ~1960s onward
Year Shown:	1967 (with Show history trail enabled)
Interactive Feature:	Year filter slider, Show history checkbox for trajectory trails

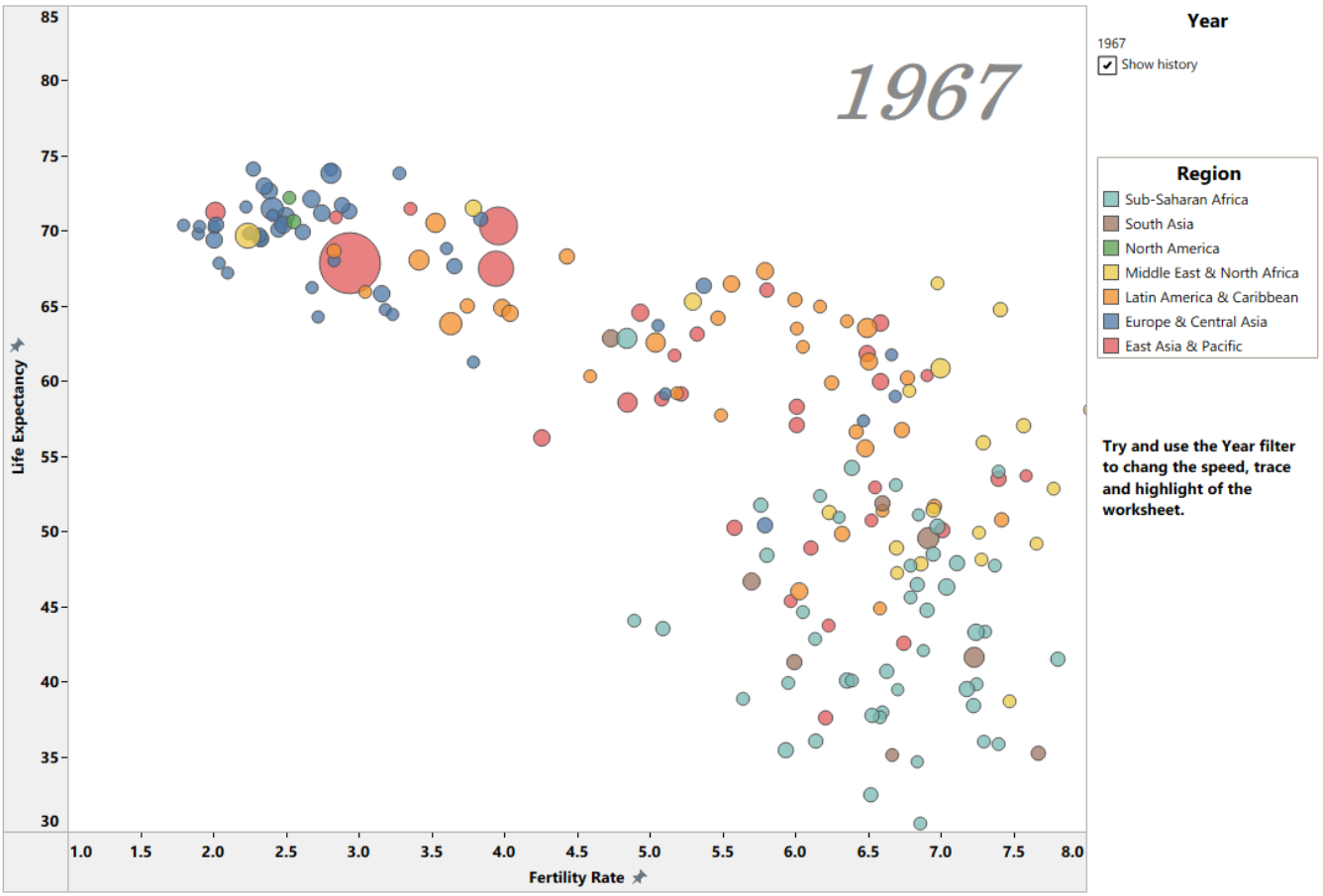
2 BUSINESS / ANALYTICAL CONTEXT

The relationship between fertility rate and life expectancy is one of the most powerful macro-level signals in global development analytics. Countries with high fertility rates (5-8 births/woman) tend to have low life expectancy (35-55 years), typically correlated with lower GDP, limited healthcare access, and high infant mortality. As countries develop economically, both metrics shift toward lower fertility and higher life expectancy -- a pattern known as the "demographic transition."

- In 1967 (view shown): Sharp negative correlation visible -- high-fertility countries (Sub-Saharan Africa, South Asia) cluster in the lower-right (high fertility, low life expectancy).
- Europe & Central Asia (blue) and North America (green): Clustered upper-left -- low fertility (1.5-3.0), high life expectancy (68-75 years).
- East Asia & Pacific (red/pink): Spans a wide range in 1967, capturing both highly developed (Japan) and developing nations (Cambodia, Laos).
- Latin America & Caribbean (orange): Mid-range fertility (4-6) with moderate life expectancy (55-70) -- transitioning economies.
- Large bubbles (China, India) visible in East Asia & Pacific and South Asia respectively, dominating by population size.
- "Show history" trails: When enabled, each bubble leaves a path showing its trajectory across previous years -- visualizing each country's development path through time.

3 DASHBOARD VIEW -- ANIMATED SCATTER: YEAR 1967

Story 2



World Life Expectancy by Fertility Rates -- Year 1967. Scatter: Fertility Rate (x) vs Life Expectancy (y). Bubble size = population. Color = 7 world regions. "Show history" enabled. Upper-left cluster: developed nations. Lower-right: high-fertility, low-life-expectancy nations.

The cluster of large orange bubbles (Latin America & Caribbean) at fertility 4-6, life expectancy 60-70 represents the demographic transition in progress in 1967 -- nations actively moving from high-fertility/low-life-expectancy toward the lower-left developed-nation cluster.

4 TECHNICAL EXPLANATION -- TABLEAU TECHNIQUES

This animated scatter plot demonstrates several advanced Tableau features working in combination:

- Pages shelf (Animation): [Year] placed on the Pages shelf creates a frame-by-frame animation. Each year value generates a "page" -- advancing the animation steps through the data chronologically. The built-in playback controls (Play, Step, Speed) are available to viewers.
- Show history: A Tableau Pages shelf option that traces each mark's movement across frames -- displaying trajectory lines/dots from past years. Configured via the History dropdown on the Pages card.
- Bubble size: SUM([Population]) on the Size marks card. Tableau automatically sizes bubbles proportionally -- China and India dominate the East Asia and South Asia regional clusters.
- Color by region: [Region] dimension on Color marks card with a 7-category qualitative palette (no sequential or diverging scale needed since regions are nominal).
- Story (Tableau Story): The scatter is embedded inside a Tableau Story as a "Story Point" -- enabling narrative captions above the viz and tab navigation if multiple story points exist.
- Interactive filter: Year slider in the view allows viewers to jump to any year directly without using the Pages playback controls.
- Axis ranges: Fixed axis ranges (Fertility: 1.0-8.0+, Life Expectancy: 30-85) lock the view across all animation frames, preventing axis rescaling that would create misleading visual jumps.

5 REGIONAL COLOR LEGEND

Sub-Saharan Africa	Teal/cyan -- largest cluster in lower-right (high fertility, low life expectancy)
South Asia	Brown/tan -- includes India (largest bubble in region)
North America	Green -- upper-left (low fertility 1.5-2.5, high life expectancy 70-75)
Middle East & North Africa	Yellow/gold -- mid-range fertility, improving life expectancy
Latin America & Caribbean	Orange -- transitioning: fertility 4-6, life expectancy 55-70
Europe & Central Asia	Blue -- upper-left cluster (developed, low fertility, high life expectancy)
East Asia & Pacific	Red/pink -- widest spread; includes Japan (upper-left) and developing nations (lower-right)

6 SKILLS DEMONSTRATED

- Pages shelf animation: Year-driven frame-by-frame playback showing demographic transition over decades.
- Show history trajectory trails: Visualizing each country's historical path through the scatter space.
- Triple mark encoding: X-axis (fertility), Y-axis (life expectancy), Size (population), Color (region) -- 4 dimensions on one mark.
- Tableau Story authoring: Embedding animated views inside a Story with narrative captions.
- Fixed axis ranges across animation frames: Ensuring consistent visual scale to prevent misleading rescaling.
- Global health dataset analysis: Connecting Gapminder-style population data to demographic transition theory.
- Interactive Year filter: Dual input (Pages playback + Year slider) for flexible viewer exploration.